

a Study design

Benchmarks 160 tasks

**BixBench-Verified-50**
50 questions · Answer
Evaluator acceptance, 0 or 1

**CompBioBench**
100 questions · Answer
Reconstructed-key agreement, 0 or 1

**IWC**
10 workflows (9 scored) · Output files
Workflow-output agreement, 0–1

Experimental design

	Custom code	Galaxy
Runs in	Isolated workspace	usegalaxy.org via MCP
Tools	Software the agent installs	Galaxy tools and UDTs*
Prompt	Task	Task + Galaxy policy
Time limit	120 min†; 6 or 12 h‡	6 or 12 h‡

*Not on IWC †CompBioBench ‡IWC; none stated otherwise

➔ **Four model configurations (Codex agent)** 3 independent runs per condition ➔

Model	Custom code	Galaxy
GPT-5.5	• • •	• • •
GPT-5.6 Sol	• • •	• • •
GPT-5.6 Luna	• • •	• • •
DeepSeek V4 Pro	• • •	• • •

$$160 \times 4 \times 2 \times 3 = 3,840$$

tasks model configs conditions runs primary runs

3,816 runs scored on 159 tasks; 4,240 archived runs include configurations outside this comparison

Evaluation

**Accuracy**
Benchmark-specific score; not pooled

**Repeatability**
Same outcome in all 3 runs?

**Input tokens**
Per run, including cached

**Galaxy interface**
Requests · failures · UDTs

**Failure causes**
AI-assisted audit of failures

b Isolated execution pipeline applied to each of the 3,840 primary runs

Galaxy condition only

